

High-Level Design (HLD) Document

Project: Gujarat Sentinel Grid — Unified CCTV Integration & Intelligence Platform
Architecture Model: Hybrid Innovative Architecture (Model 5)

1. Executive Summary

The **Gujarat Sentinel Grid** provides an open-source, vendor-neutral platform for law enforcement agencies to ingest, analyze, and correlate live CCTV video feeds across 26 government departments and private entities. Built to solve geographical dispersion (up to 1,000 km) and heterogeneous infrastructure (mixed VMS, H.264/H.265 codecs, analog/IP), the platform features:

- **Dynamic Catalogue Discovery** via `http://<host>/api/ingest` (zero hardcoded endpoints).
- **Multi-Protocol Video Ingestion** (WebRTC WHEP for low-latency UI, RTSP over TCP for AI processing).
- **Automatic Number Plate Recognition (ANPR)** and real-time Watchlist Database Correlation (VAHAN, eGujCop).
- **Vehicle Trajectory & Route Reconstruction Engine** for hackathon evaluation test cases.

2. Technology Stack Justification

Component	Technology	Rationale
Frontend UI	React, Tailwind CSS, Leaflet GIS	Open-source, responsive, high-performance spatial map rendering.
Video Protocols	WebRTC WHEP, RTSP over TCP	WebRTC delivers <200ms preview latency; RTSP over TCP guarantees zero packet loss for AI workers.
Backend & APIs	Python FastAPI, Uvicorn	Asynchronous execution, native integration with OpenCV/PyTorch models.
AI Inference	YOLOv8, FastALPR, OpenCV	High-speed vehicle detection and license plate extraction.
Spatial Database	PostgreSQL + PostGIS	Enterprise spatial indexing for instant vehicle route reconstruction.

3. Key Security & Compliance Rules

1. **RTSP Transport Enforcement:** All workers force `rtsp_transport=tcp` to eliminate UDP corruption across NAT/firewalls.
2. **PTS Timestamping:** Timing calculations rely strictly on Presentation Timestamps (`CAP_PROP_POS_MSEC`), avoiding variable FPS inaccuracies.
3. **Resilient Reconnection:** Exponential backoff reconnects (2s to 30s) prevent gateway flood risks during network interruptions.

4. **Role-Based Access Control:** Department-wise permissioning ensures data isolation across civil supplies, transport, and police domains.

Official GitHub Repository: <https://github.com/AREKG0/cctv-integration-hackathon>